

AI MINI PROJECT

PawSense AI – Dog Behaviour Analysis

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2. Introduction :

PawSense AI is an AI-based system that analyzes commonly observed dog behaviours and provides a concern-level result. The system uses a Decision Tree Machine Learning algorithm to classify the observed behaviour into Normal, Mild Concern, Moderate Concern, or High Concern. The project is useful because it provides a simple way to observe unusual behaviour and get an AI-based indication. It is an educational support system and does not provide medical diagnosis.

3. Problem Statement :

Dog owners may notice changes in a dog's behaviour but may not know how to interpret multiple signs together. PawSense AI aims to provide a simple AI-based system that analyzes selected dog behaviours and generates a concern-level result for basic awareness and observation.

4. Objectives

- To collect and prepare a dog behaviour dataset.
- To preprocess the dataset for Machine Learning.
- To apply a Decision Tree algorithm.
- To train and test the AI model.
- To generate concern-level predictions based on selected behaviours.

5. Technologies Used :

- Python
- Flask
- Pandas
- NumPy
- Scikit-learn
- HTML
- CSS
- JavaScript
- GitHub
- Vercel

6. Dataset

Dataset Name: PawSense AI Behaviour Dataset

Source: Self-created synthetic/educational dataset.

Number of Records: 50

Features:

- Limping
- Low Activity
- Paw Licking
- Scratching
- Eating Less
- Unusual Sounds

Target Variable: Concern

The target variable contains four categories:

- Normal
- Mild Concern
- Moderate Concern
- High Concern

7. Methodology / Working

1. Create and prepare the dog behaviour dataset.
2. Load the dataset using Pandas.
3. Separate input features and target variables.
4. Split the dataset into training and testing data.
5. Train the Decision Tree Classifier.
6. Test the trained model and calculate accuracy.
7. Save the trained model using Pickle.
8. Take behaviour inputs from the user through the web interface.
9. Send the inputs to the Flask backend.
10. Generate the prediction and confidence.
11. Display the result, explanation, guidance, history and trend.

8. Algorithm Used :

A **Decision Tree** is a Machine Learning algorithm that makes predictions by following a tree-like structure of decisions.

In PawSense AI, the algorithm checks the selected behaviour features and follows decision rules to classify the dog behaviour into a concern category.

Why is it used?

- Easy to understand.
- Suitable for classification.

- Works well with simple yes/no features.
- Easy to explain during viva.

How does it Works:

Behaviour Inputs



Decision Tree



Check Behaviour Conditions



Concern Category



Normal / Mild / Moderate / High

9. Implementation :

The project is implemented using Python, Flask and Scikit-learn.

Important implementation parts include:

Dataset Loading

```
data = pd.read_csv("dataset.csv")
```

This loads the PawSense AI dataset.

Model Creation

```
model = DecisionTreeClassifier(
    max_depth=4,
    random_state=42
)
```

This creates the Decision Tree model.

Model Training

```
model.fit(X_train, y_train)
```

The model learns patterns from the training data.

Prediction

```
prediction = model.predict([features])[0]
```

The trained model predicts the concern category based on user-selected behaviours.

Confidence

```
probabilities = model.predict_proba([features])[0]
confidence = round(max(probabilities) * 100, 2)
```

This displays the model's highest predicted class probability as a confidence value.

10. Results :

Show the final results of the project. Include:

- Accuracy/performance
- Prediction results
- Graphs
- Output screenshots

11. Advantages :

- Simple and easy-to-use interface.
- Uses AI for behaviour classification.
- Provides quick concern-level results.
- Shows prediction confidence.
- Maintains behaviour history and trend.
- Can be accessed through a web browser.

12. Limitations :

- Dataset contains only 50 synthetic records.
- The system does not provide medical diagnosis.
- Accuracy may change with a larger real-world dataset.
- It depends on the behaviours selected by the user.

13. Applications :

- Basic dog behaviour observation.
- Educational AI/ML demonstration.
- Pet-owner awareness.
- Veterinary-support information before consultation.
- Demonstration of Machine Learning classification in a web application.

14. Conclusion :

PawSense AI is an AI-based dog behaviour observation system developed using Python, Flask and Decision Tree Machine Learning. The system analyzes selected behaviour inputs and generates a concern-level prediction along with confidence and basic guidance.

The project demonstrates how AI and Machine Learning can be integrated into a simple web application for practical use and awareness.

15. Future Scope :

In the future, the project can be improved by:

- Adding a larger real-world dataset.
- Improving model accuracy.
- Adding more behaviour features.
- Adding user accounts and cloud-based history.
- Developing a mobile application.
- Adding image-based dog behaviour analysis.
- Adding multiple animals or birds behavior analysis

16. References :

Website Link: <https://pawsense-ai.vercel.app/>

GitHub Link: <https://github.com/radhikamali18/PawSense-AI>